

Name _____



2022

Year 9 Level 5.3
Assessment Task 4 (AT4)
November 2022

Mathematics

General Instructions

- Working time – 90 minutes
- Calculators approved by NESA may be used
- Marks for each question are indicated on the question paper
- Write using a black pen

Total marks: 77

Section I 10 marks

- Attempt Questions 1-10
- Use the Multiple Choice Answer Sheet provided
- Allow about 10 minutes for this section

Section II 67 marks

- Answer Questions 11-15 in the spaces provided
- In Questions 11-15, show relevant mathematical reasoning and/or calculations
- Allow about 80 minutes for this section

Examiners' Use Only				Setter/Checker: ETR/MAS		
Question	Algebra	Geometry	Measurement	Number	Total	
Section I						
1-10	3, 4 / 2	5, 6, 7 / 3	8, 9, 10 / 3	1, 2 / 2	/ 10	
Section II						
11			/ 15		/15	
12		a / 5		b, c /5	/10	
13	c, d / 8			a, b / 6	/14	
14	/ 14				/14	
15	/ 14				/14	
Total	/ 38	/ 8	/ 18	/13	/77	

Section I

10 marks

Attempt Questions 1-10

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

1 13 364 written in scientific notation, correct to three significant figures, is:

A. 13 300

B. 1.33×10^4

C. 1.34×10^4

D. 1.33×10^5

2 Given $\sqrt{y} = 5\sqrt{6}$, the value of y is:

A. 5

B. 6

C. 30

D. 150

3 What is the gradient for the equation $2x + 4y = 10$?

A. -2

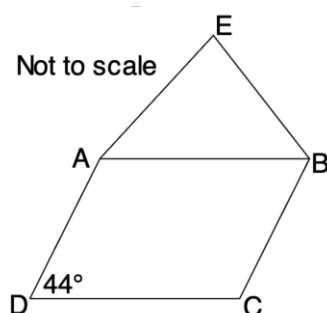
B. -0.5

C. 0.5

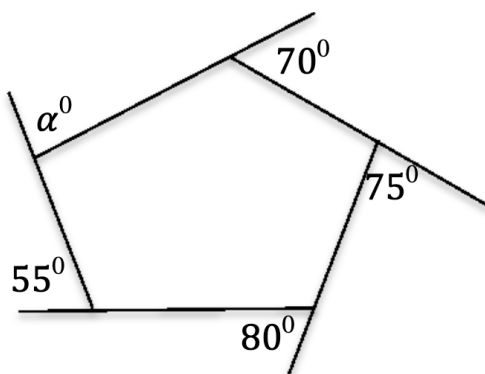
D. 2

- 4 The product of a number, and 2 less than the same number, is 8. What could be the original number?
- A. -2
- B. 0
- C. 2
- D. 5

- 5 $ABCD$ is a rhombus with $\angle ADC = 44^\circ$. $\triangle ABE$ is an equilateral triangle. The size of $\angle EAD$ is:

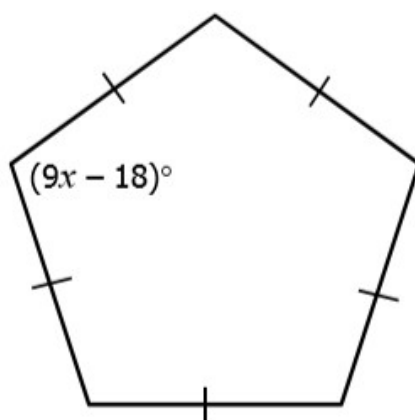


- A. 44°
- B. 60°
- C. 104°
- D. 196°
- 6 The size angle of angle α° is:

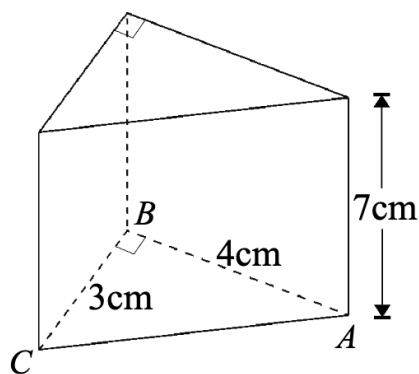


- A. 70°
- B. 75°
- C. 80°
- D. 95°

- 7 The diagram below shows a regular pentagon. What is the value of x ?

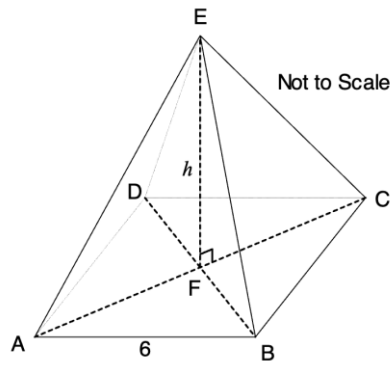


- A. 10
B. 12
C. 14
D. 16
- 8 What is the surface area of the shape below?



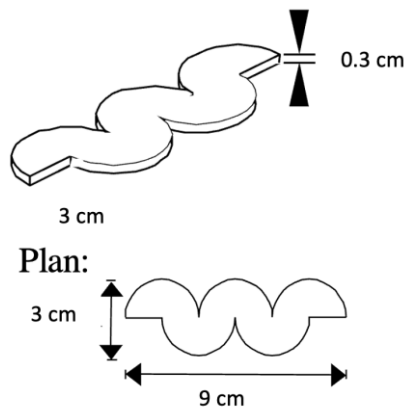
- A. 42 cm^2
B. 61 cm^2
C. 96 cm^2
D. 108 cm^2

- 9 A pyramid with a square base has $AB = 6$ cm and $EF = h$ cm. EC is $\sqrt{82}$ cm.



The value of h is:

- A. 5
 - B. 6
 - C. 7
 - D. 8
- 10 Badges have been designed as shown below.



All the curves in the design are semi-circles with the same radius. The thickness is 0.3 cm. The volume of the badge (to the nearest 1 decimal place) is:

- A. 5.3 cm^3
- B. 10.6 cm^3
- C. 21.2 cm^3
- D. 42.4 cm^3

Name _____

Section II

67 marks

Attempt Questions 11-15

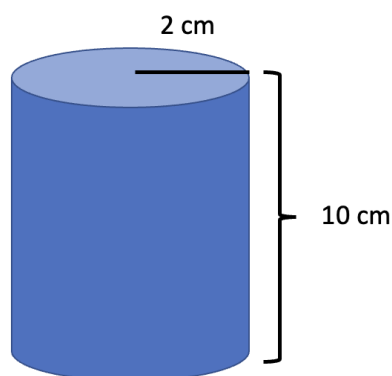
Allow about 80 minutes for this section

- Answer the questions in the spaces provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing paper will be provided by your teacher if required.

Question 11 (Measurement) (15 marks)

Marks

- (a) A closed cylinder has a height of 10 cm and a radius of 2 cm.



- (i) What is the volume of the cylinder? Answer to 2 decimal places.

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- (ii) What is the surface area of the cylinder? Answer to 2 decimal places.

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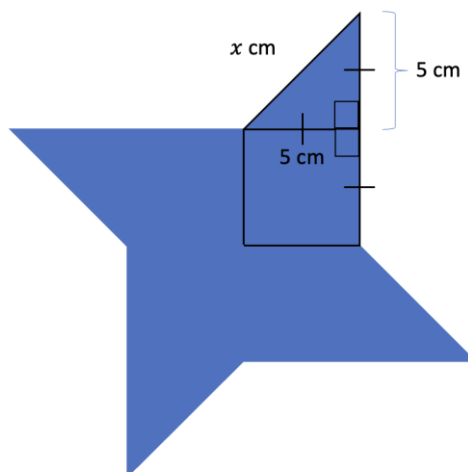
Question 11 continues on the next page

Name

Question 11 (continued)

Marks

- (b) Naruto constructs the shuriken for his work. It is made of sections that are all the same.



- (i) Find the value of x . Answer to 2 decimal places.

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- (ii) Find the area of one side of the shuriken.

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Question 11 continues on the next page

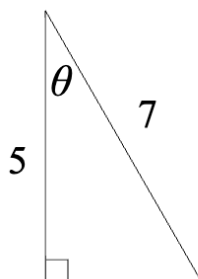
Name

Question 11 (continued)

Marks

- (c) Find the value of θ . Give your answer to the nearest degree.

2



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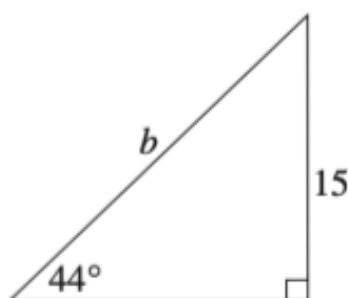
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- (d) Find the value of b . Give your answer to the nearest 2 decimal places.
All lengths are in cm.

2



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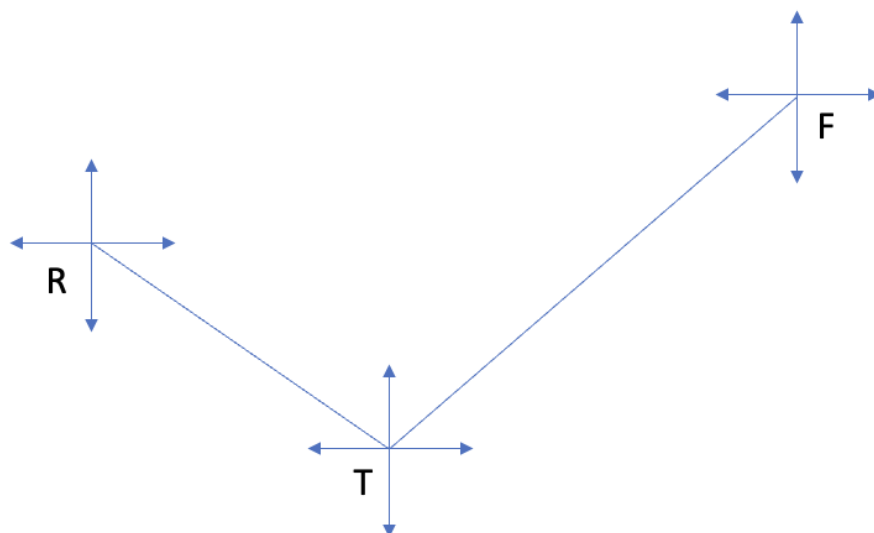
Question 11 continues on the next page

Name

Question 11 (continued)

Marks

- (e) Rapunzel (R) and Flynn (F) both leave from a tower (T). Rapunzel walks on a bearing of 320° for 5 km. Flynn walks on a bearing of 50° for 10 km.



- (i) Show that $\angle RTF = 90^\circ$. **1**

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- (ii) Find the distance between Rapunzel and Flynn. Leave your answer in exact form. **1**

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- (iii) What is the bearing of Rapunzel from Flynn? Answer to the nearest degree. **2**

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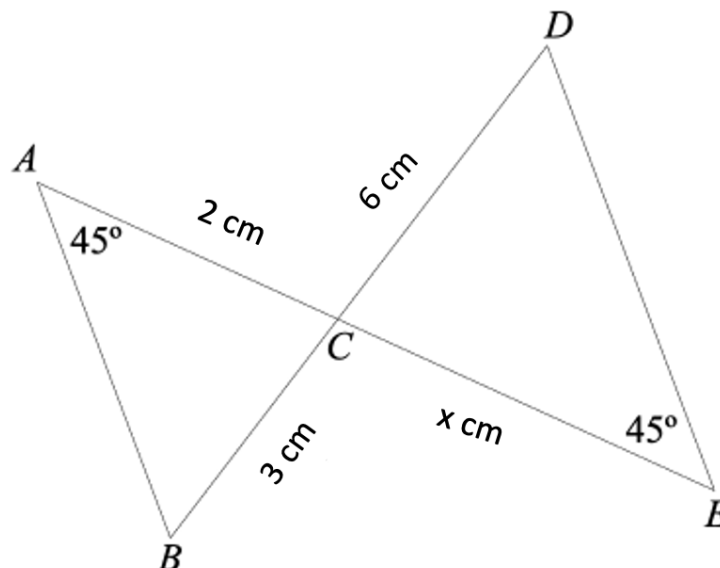
End of Question 11

Name

Question 12 (Geometry and Number) (10 marks)

Marks

- (a) In the diagram below, C lies on the lines AE and BD .



- (i) Prove that triangle ABC is similar to triangle EDC , giving clear reasons.

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- (ii) Find the value of x . You must give a reason.

2

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Question 12 continues on the next page

Name

Question 12 (continued) **Marks**

- (b) Mr Burns wants to save enough money to open a nuclear powerplant in Springfield. **2**
 If he invests \$1500 into a bank at 10 % p.a compounded yearly for 15 years, how much money will Mr Burns have in total? Answer to 2 decimal places.

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- (c) The following table shows the Australian tax rates for individuals.

Taxable income	Tax on this income
\$0 – \$18 200	Nil
\$18 201 – \$37 000	19c for each \$1 over \$18 200
\$37 001 – \$87 000	\$3572 plus 32.5c for each \$1 over \$37 000
\$87 001 – \$180 000	\$19 822 plus 37c for each \$1 over \$87 000
\$180 001 and over	\$54 232 plus 45c for each \$1 over \$180 000

- (i) Homer works at the local nuclear powerplant. He made \$95000 in the year. **1**
 Homer also donated \$2000 to the local charity and had \$1000 in work-related expenses. What is Homer’s taxable income?

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- (ii) Using the table above, how much tax does Homer need to pay? **2**

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End of Question 12

Name

Question 13 (Number and Algebra) (14 marks)

Marks

(a) Simplify the following:

(i) $2\sqrt{3} + 3\sqrt{3}$ **1**

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(ii) $2\sqrt{5} \times 3\sqrt{4}$ **2**

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(b) Simplify $\frac{(a^{-4})^3 b^4}{a^9 (b^{-1})^3}$ and write your answer with positive indices: **3**

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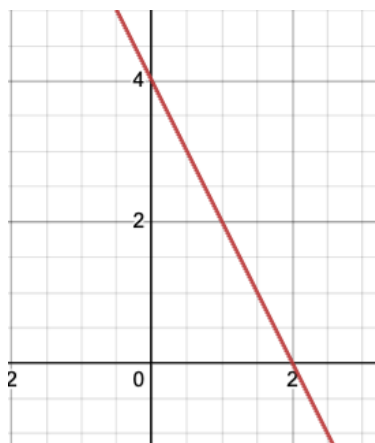
Question 13 continues on the next page

Name _____

Question 13 (continued)

Marks

(c) Dr Seuss was showing his students the line below.



(i) What is the gradient of the line? **1**

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(ii) What is the y -intercept of the line? **1**

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(iii) What is the equation of the line? **1**

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(iv) Does the point $(2, 2)$ lie on the line? **1**

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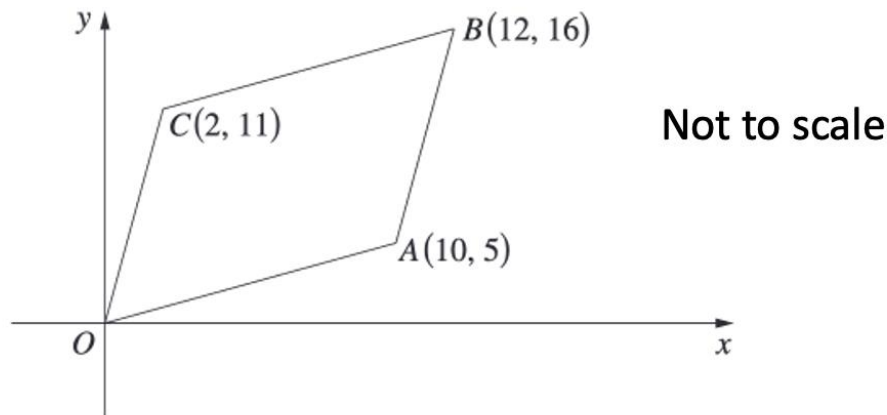
Question 13 continues on the next page

Name

Question 13 (continued)

Marks

- (d) In the diagram below, A , B and C are the points $(10, 5)$, $(12, 16)$ and $(2, 11)$ respectively.



- (i) Calculate the distance AC . 1

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- (ii) Calculate the midpoint of AC . 1

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- (iii) Show that OB and AC are perpendicular. 2

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End of Question 13

Name _____

Question 14 (Algebra) (14 marks)

Marks

(a) Solve the following equations:

(i) $2x - 1 = 11$ **1**

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(ii) $x^2 - 9x = 0$ **2**

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(iii) $2x^2 + 5x + 2 = 0$ **2**

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(b) Fully factorise the following:

(i) $2x^2 - x$ **1**

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(ii) $ax^2 + ax + 2x + 2$ **2**

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Question 14 continues on the next page

Name _____

Question 14 (continued)

Marks

(c) Fully simplify the following:

(i) $\frac{32}{2a+6} \div \frac{16}{(a+3)^2}$

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(ii) $\frac{12}{x^2-16} - \frac{3}{x+4}$

3

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End of Question 14

Name

Question 15 (Algebra) (14 marks)

Marks

(a) Sketch $y = x^2$ and $y = 3x^2$ on the same set of axes.

2

(b) Sketch $y = (x - 1)^2 + 4$, showing the vertex, axis of symmetry and y -intercept.

4

Question 15 continues on the next page

Name

Question 15 (continued)

Marks

(c) A parabola has the equation: $y = x^2 - 4x + 3$

(i) Show that the x -intercepts for the parabola are at $x = 3$ and $x = 1$. **2**

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(ii) Find the y -intercept of the parabola. **1**

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(iii) Sketch the parabola, showing the x -intercepts and y -intercept. **2**

Question 15 continues on the next page

Name

Question 15 (continued)

Marks

- (d) Hermione was on the phone trying to describe a parabola to Harry. However, Hermione had bad reception and parts of the conversation had been cut out. Hermione says that “the $y = x^2$ has been shifted _____ by 16 units”. She also says, “the x -intercepts cuts at _____ and _____”. Harry didn’t hear the exact x -intercepts but knew that **one** of the x -intercepts was between 2 and 4 (inclusive of 2 and 4). The call then ended.

Given that the equation of a parabola is $y = a(x - h)^2 + c$, find the possible value(s) of a, h and c .

[illegible]

End of Question 15

End of Assessment Task

Name _____



THE
KING'S
SCHOOL

2022

Year 9 Level 5.3
Assessment Task 4 (AT4)
November 2022

MULTIPLE CHOICE ANSWER SHEET

Select the alternative A, B, C or D that best answers the question and indicate your choice by filling in the appropriate response oval, completely.

					<i>Examiner's Use Only</i>	
1	(A)	(B)	(C)	(D)		
2	(A)	(B)	(C)	(D)	Number	/ 2
3	(A)	(B)	(C)	(D)		
4	(A)	(B)	(C)	(D)	Algebra	/ 2
5	(A)	(B)	(C)	(D)		
6	(A)	(B)	(C)	(D)		
7	(A)	(B)	(C)	(D)	Geometry	/ 3
8	(A)	(B)	(C)	(D)		
9	(A)	(B)	(C)	(D)		
10	(A)	(B)	(C)	(D)	Measurement	/ 3
					Total	/10

Name _____

Name _____



2022

Year 9 Level 5.3
Assessment Task 4 (AT4)
November 2022

Mathematics

- General Instructions
- Working time – 90 minutes
 - Calculators approved by NESA may be used
 - Marks for each question are indicated on the question paper
 - Write using a black pen

- Total marks: 77
- Section I 10 marks

 - Attempt Questions 1-10
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Section II 67 marks

 - Answer Questions 11-15 in the spaces provided
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Examiners' Use Only					Setter/Checker: ETR/MAS				
Question	Algebra		Geometry		Measurement		Number		Total
Section I									
1-10	3, 4	/ 2	5, 6, 7	/ 3	8, 9, 10	/ 3	1, 2	/ 2	/ 10
Section II									
11					/ 15				/15
12			a	/ 5			b, c	/5	/10
13	c, d	/ 8					a, b	/ 6	/14
14		/ 14							/14
15		/ 14							/14
Total		/ 38		/ 8		/ 18		/13	/77

Section I

10 marks

Attempt Questions 1-10

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

1 13 364 written in scientific notation, correct to three significant figures, is:

A. 13 300

B. 1.33×10^4

☒ C. 1.34×10^4

D. 1.33×10^5

$$\underline{\underline{1.3364 \times 10^4}}$$

$$1.34 \times 10^4$$

(C)

2 Given $\sqrt{y} = 5\sqrt{6}$, the value of y is:

A. 5

B. 6

C. 30

☒ D. 150

$$\sqrt{y} = \sqrt{25 \times 6}$$

$$= \sqrt{150}$$

$$\therefore y = 150$$

(D)

3 What is the gradient for the equation $2x + 4y = 10$?

A. -2

☒ B. -0.5

C. 0.5

D. 2

$$4y = 10 - 2x$$

$$y = \frac{10}{4} - \frac{2}{4}x$$

$$y = 2.5 - \underline{0.5x}$$

$\therefore$ (B)

- 4 The product of a number, and 2 less than the same number, is 8. What could be the original number?

A. -2

B. 0

C. 2

D. 5

$$x(x-2) = 8$$

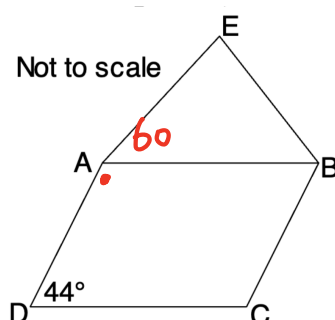
$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$$\therefore x = 4, -2$$

C

- 5 ABCD is a rhombus with $\angle ADC = 44^\circ$. $\triangle ABE$ is an equilateral triangle. The size of $\angle EAD$ is:



$$= 180 - 44$$

$$= 136$$

$$\therefore 136 + 60$$

$$= 196^\circ$$

A. 44°

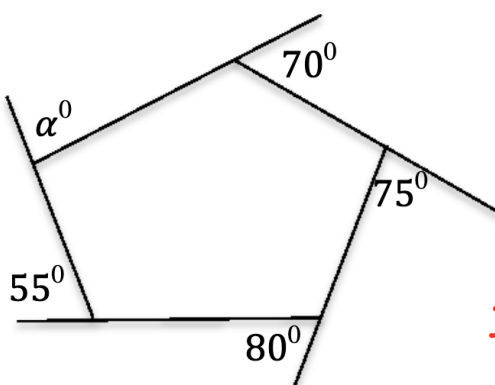
B. 60°

C. 104°

D. 196°

D

- 6 The size angle of angle α° is:



$$360^\circ = \text{total ext } \angle$$

$$\therefore \alpha = 360 - 70 - 75$$

$$- 80 -$$

$$55$$

$$= 80^\circ$$

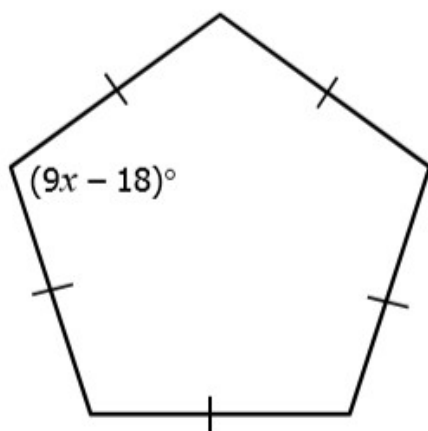
A. 70°

B. 75°

C. 80°

D. 95°

- 7 The diagram below shows a regular pentagon. What is the value of x ?



A. 10

B. 12

C. 14

D. 16

$$\frac{180(n-2)}{n} = \text{each } \angle^{\text{int}}$$

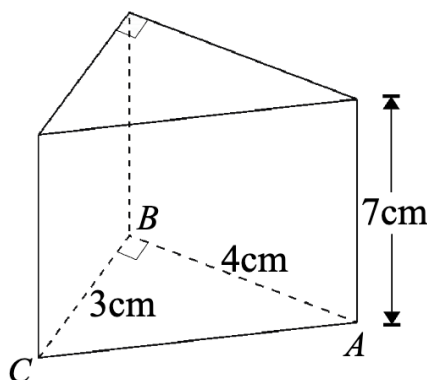
$$n = 5, \therefore 108^\circ$$

$$9x - 18 = 108$$

$$9x = 126, x = \frac{126}{9} = 14$$

(C)

- 8 What is the surface area of the shape below?



$$AC = \sqrt{3^2 + 4^2} = 5$$

A. 42 cm²

B. 61 cm²

C. 96 cm²

D. 108 cm²

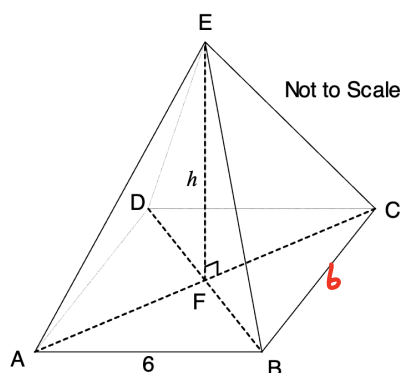
$$2 \times \left(3 \times 4 \times \frac{1}{2} \right) + 3 \times 7 + 7 \times 4 + 5 \times 7$$

$$2\Delta + \square + \square + \square$$

$$= 96 \text{ cm}^2$$

(C)

- 9 A pyramid with a square base has $AB = 6$ cm and $EF = h$ cm. EC is $\sqrt{82}$ cm.



The value of h is:

- A. 5
B. 6
C. 7

D. 8

$$AC = \sqrt{6^2 + 6^2}$$

$$AC = \sqrt{72}$$

$$FC = \frac{1}{2} AC = \frac{1}{2} \sqrt{72}$$

$$EF^2 + FC^2 = EC^2$$

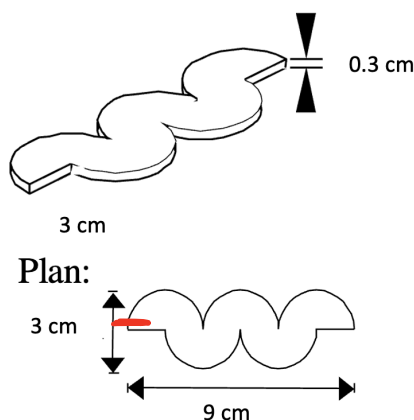
$$h^2 + \frac{1}{4}(72) = 82$$

$$h^2 = 64$$

$$h = 8$$

D

- 10 Badges have been designed as shown below.



$$\text{Diameter} = 3 \text{ cm}$$

$$r = 1.5 \text{ cm}$$

All the curves in the design are semi-circles with the same radius. The thickness is 0.3 cm. The volume of the badge (to the nearest 1 decimal place) is:

A. 5.3 cm³

B. 10.6 cm³

C. 21.2 cm³

D. 42.4 cm³

$$5 \text{ semi circles} = 5 \times \left(\frac{1}{2} \pi r^2 \right)$$

$$= 17.67$$

$$\text{Vol} = 17.67 \times \text{thickness}$$

$$= 5.3 \text{ cm}^3$$

A

Name

Section II

75 marks

Attempt Questions 11-16

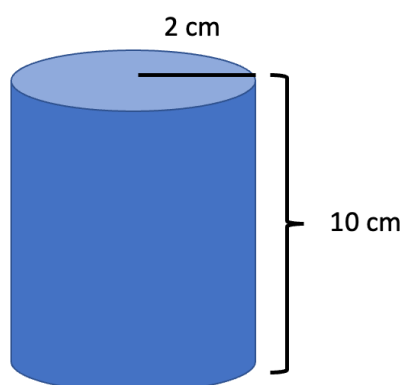
Allow about 80 minutes for this section

- Answer the questions in the spaces provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing paper will be provided by your teacher if required.

Question 11 (Measurement) (15 marks)

Marks

- (a) A closed cylinder has a height of 10 cm and a radius of 2 cm.



- (i) What is the volume of the cylinder? Answer to 2 decimal places.

2

1m formula: $V = \pi r^2 h$

1m answer: $V = \pi \times 2^2 \times 10$

$$= 125.66 \text{ cm}^3$$

- (ii) What is the surface area of the cylinder? Answer to 2 decimal places.

2

1m formula: $SA = 2\pi r^2 + 2\pi r h$

$$= 2 \times \pi \times 4 + 2 \times \pi \times 2 \times 10$$

1m answer:

$$= 150.80 \text{ cm}^2$$

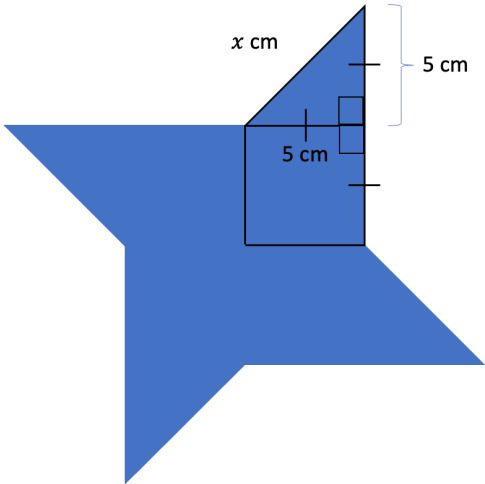
Question 11 continues on the next page

Name _____

Question 11 (continued)

Marks

(b) Naruto constructs the shuriken for his work. It is made sections that are all the same.



(i) Find the value of x . Answer to 2 decimal places. 1

Pythag: $x = \sqrt{5^2 + 5^2}$
 $= 7.07 \text{ cm}$

(ii) Find the total area of the shuriken. 2

1m for
a component found. $\triangle = 5 \times 5 \times \frac{1}{2} = 12.5 \text{ cm}^2$
 $\square = 5 \times 5 = 25$
 $\therefore \text{shuriken} = 25 + 12.5 = 37.5$

1m for final ans = $37.5 \times 4 = 150 \text{ cm}^2$

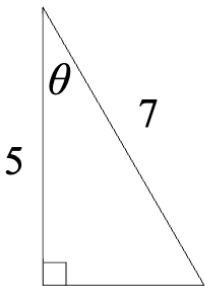
Question 11 continues on the next page

Name _____

Question 11 (continued)

Marks

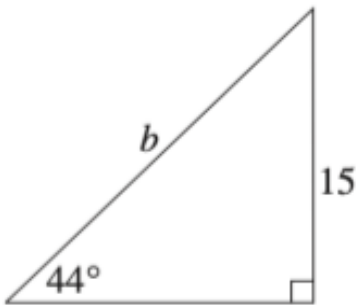
(c) Find the value of θ . Give your answer to the nearest degree. 2



Im for correct : $\cos \theta = \frac{5}{7}$
trig ratio

Im answer : $\theta = \cos^{-1}\left(\frac{5}{7}\right) = 44^\circ$

(d) Find the value of b . Give your answer to the nearest 2 decimal places. 2



Im correct trig ratio: $\sin 44 = \frac{15}{b}$

Im correct ans: $b = \frac{15}{\sin 44} = 21.59 \text{ cm}$

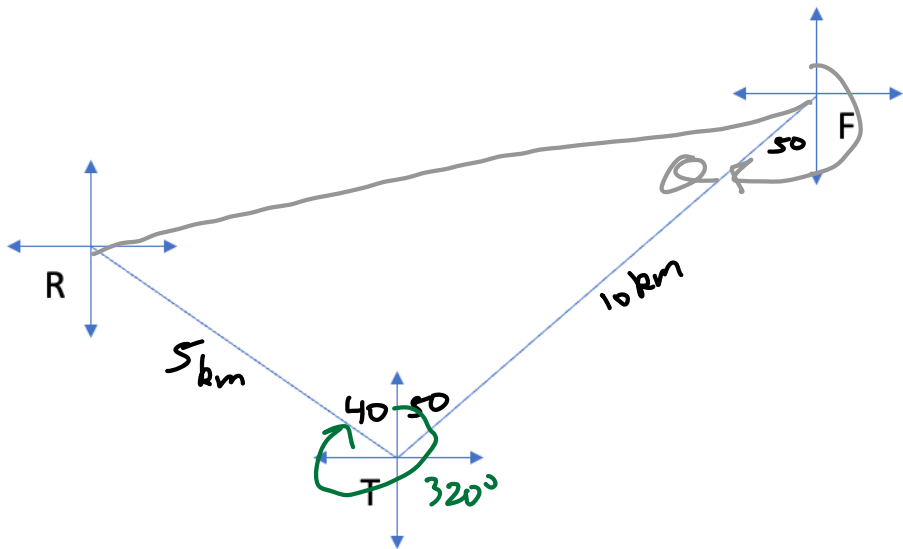
Question 11 continues on the next page

Name _____

Question 11 (continued)

Marks

(d) Rapunzel (R) and Flynn (F) both leave from a tower (T). Rapunzel walks on a bearing of 320° for 5 km. Flynn walks on a bearing of 50° for 10 km.



(i) Show that $\angle RTF = 90^\circ$. 1

1m for logical reasoning
Using the diagram, bearing of 320° means
 $360 - 320 = 40^\circ$ left. $\therefore 40 + 50$ [Flynn's bearing]
 $= 90^\circ$

(ii) Find the distance between Rapunzel and Flynn. Leave your answer in exact form. 1

1m for
correct ans : $\sqrt{5^2 + 10^2} = \sqrt{125}$ km

(iii) What is the bearing of Rapunzel from Flynn? Answer to the nearest degree. 2

Flynn to Rapunzel = $180 + 50 + \theta$
 $\tan \theta = \frac{5}{10}$
 $\theta = 27^\circ$
1m for θ
 $= 180 + 50 + 27$
 $= 257^\circ$ 1m for correct final answer.

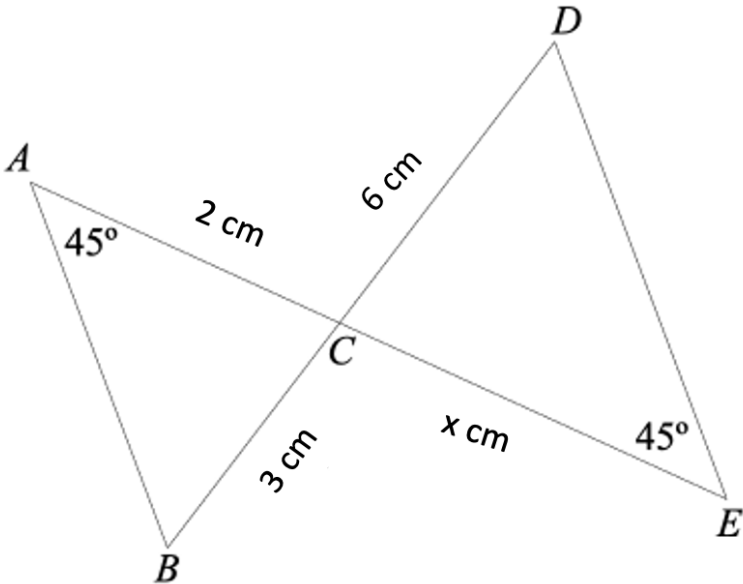
End of Question 11

Name _____

Question 12 (Geometry and Number) (15 marks)

Marks

(a) In the diagram below, C lies on the lines AE and BD .



(i) Prove that triangle ABC is similar to triangle EDC , giving clear reasons. 3

In $\triangle ABC$ & $\triangle EDC$
 $\angle BAC = \angle CED = 45^\circ$ (given) 1m
 $\angle ACB = \angle ECD$ [vert opp $\angle$ s are equal] 1m
 $\therefore \triangle ABC \parallel \triangle EDC$ [2 angles equal] 1m

(ii) Find the value of x . You must give a reason. 2

$\frac{x}{2} = \frac{6}{3}$ [similar $\triangle$ s have sides in ratio] 1m

$\therefore x = 4 \text{ cm}$

1m for
1m for answer reason.

Question 12 continues on the next page

Name _____

Question 12 (continued)

Marks

- (b) Mr Burns wants to save enough money to open a nuclear powerplant in Springfield. 2
- If he invests \$1500 into a bank at 10 % p.a compounded yearly for 15 years, how much money will Mr Burns have in total? Answer to 2 decimal places.

Im for formula: $A = P(1 + r)^n$

Im for ans: $A = 1500(1 + 10\%)^{15}$

$= \$6265.87$

- (c) The following table shows the Australian tax rates for individuals.

Taxable income	Tax on this income
\$0 – \$18 200	Nil
\$18 201 – \$37 000	19c for each \$1 over \$18 200
\$37 001 – \$87 000	\$3572 plus 32.5c for each \$1 over \$37 000
\$87 001 – \$180 000	\$19 822 plus 37c for each \$1 over \$87 000
\$180 001 and over	\$54 232 plus 45c for each \$1 over \$180 000

- (i) Homer works at the local nuclear powerplant. He made \$95000 in the year. 1
- Homer also donated \$2000 to the local charity and had \$1000 in work-related expenses. What is Homer’s taxable income?

$95000 - 2000 - 1000 = \$92000$

- (ii) Using the table above, how much tax does Homer need to pay? 2

Im -or correct row: $19822 + 0.37(92000 - 87000)$

Im for final answer $= \$21672$

End of Question 12

Name _____

Question 13 (Number and Algebra) (14 marks)

Marks

(a) Simplify the following:

(i) $2\sqrt{3} + 3\sqrt{3}$

1

$5\sqrt{3}$

(ii) $2\sqrt{5} \times 3\sqrt{4}$

2

Im for correct multiplication : $6\sqrt{20}$

Im for simplifying : $6 \times \sqrt{4} \times \sqrt{5}$

$= 6 \times 2 \times \sqrt{5}$

$= 12\sqrt{5}$

(b) Simplify $\frac{(a^{-4})^3 b^4}{a^9 (b^{-1})^3}$ and write your answer with positive indices:

3

Im for correct power of power: $\frac{a^{-12} b^4}{a^9 b^{-3}}$

Im for positive indices: $\frac{b^3 b^4}{a^9 a^{12}}$

Im for simplified: $\frac{b^7}{a^{21}}$

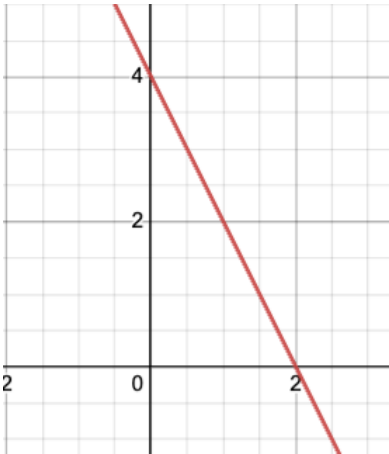
Question 13 continues on the next page

Name _____

Question 13 (continued)

Marks

(c) Dr Seuss was showing his students the line below.



(i) What is the gradient of the line? 1

$$m = \frac{-4}{2} = \boxed{-2}$$

(ii) What is the y-intercept of the line? 1

$$b = \boxed{4}$$

(iii) What is the equation of the line? 1

$$y = mx + b$$

$$\boxed{y = -2x + 4}$$

(iv) Does the point (2, 2) lie on the line? 1

From the graph, (2, 2) does NOT lie on the line.

OR

$$y = -2x + 4$$

$$LHS = 2, \quad RHS = -2(2) + 4 = 0 \quad \therefore L \neq R$$

$$\therefore \text{Not lie}$$

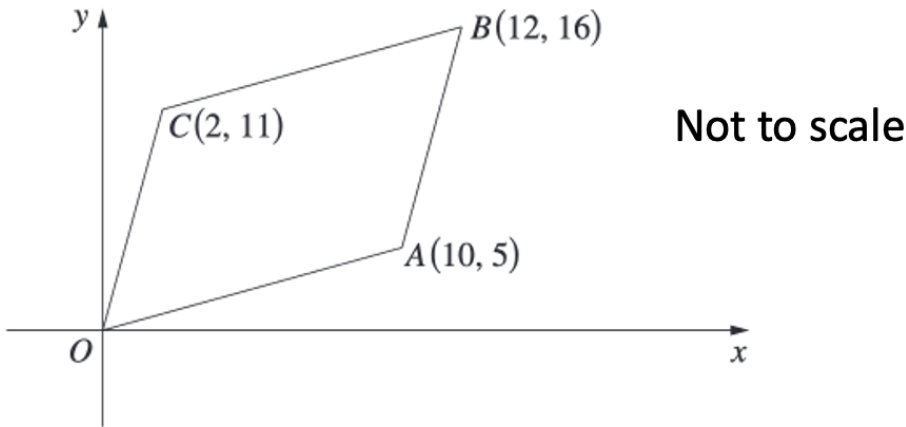
Question 13 continues on the next page

Name _____

Question 13 (continued)

Marks

(d) In the diagram below, A , B and C are the points $(10, 5)$, $(12, 16)$ and $(2, 11)$ respectively.



(i) Calculate the distance AC . 1

$$AC = \sqrt{(10 - 2)^2 + (5 - 11)^2}$$
$$= \boxed{10 \text{ units}}$$

(ii) Calculate the midpoint of AC . 1

$$\text{Midpoint } AC = \left(\frac{2+10}{2}, \frac{11+5}{2} \right) = (6, 8)$$

(iii) Show that OB and AC are perpendicular. 2

$$m_{OB} = \frac{16 - 0}{12 - 0} = \frac{4}{3} \qquad m_{AC} = \frac{11 - 5}{2 - 10} = -\frac{2}{4}$$

$$m_{OB} \times m_{AC} = \frac{4}{3} \times -\frac{2}{4} = -1$$
$$\therefore OB \perp AC$$

Im for m_{OB} or m_{AC}

Im for $m_{OB} \times m_{AC} = -1$

End of Question 13

Name _____

Question 14 (Algebra) (14 marks)

Marks

(a) Solve the following equations:

(i) $2x - 1 = 11$

1

$2x = 12$

$x = 6$

(i) $x^2 - 9x = 0$

2

$x(x - 9) = 0$

$\therefore x = 0, x = 9$ 1m per answer

(ii) $2x^2 + 5x + 2 = 0$

2

$2x \quad x \quad 1 \quad 2$

$\therefore (2x + 1)(x + 2)$

$x = -\frac{1}{2}, x = -2$ 1m per answer.

(b) Fully factorise the following:

(i) $2x^2 - x$

1

$x(2x - 1)$

(ii) $ax^2 + ax + 2x + 2$

2

$a(x + 1) + 2(x + 1)$ 1m

$(x + 1)(ax + 2)$ 1m

Question 14 continues on the next page

Name _____

Question 14 (continued)

Marks

(c) Fully simplify the following:

(i) $\frac{32}{2a+6} \div \frac{16}{(a+3)^2}$

3

Im for doing $\div$ correctly: $\frac{32}{2a+6} \times \frac{(a+3)^2}{16}$

Im for factorise: $\frac{\cancel{32}}{\cancel{2(a+3)}} \times \frac{(a+3)^2}{\cancel{16}}$

Im for simplified: $\boxed{a+3}$

(ii) $\frac{12}{x^2-16} - \frac{3}{x+4}$

3

Im for diff of squares: $\frac{12}{(x-4)(x+4)} - \frac{3}{(x+4)}$

Im for same denom: $\frac{12}{(x-4)(x+4)} - \frac{3(x-4)}{(x+4)(x-4)}$

Im fully simplified: $\frac{12 - 3x + 12}{(x-4)(x+4)} = \boxed{\frac{-3x}{(x-4)(x+4)}}$

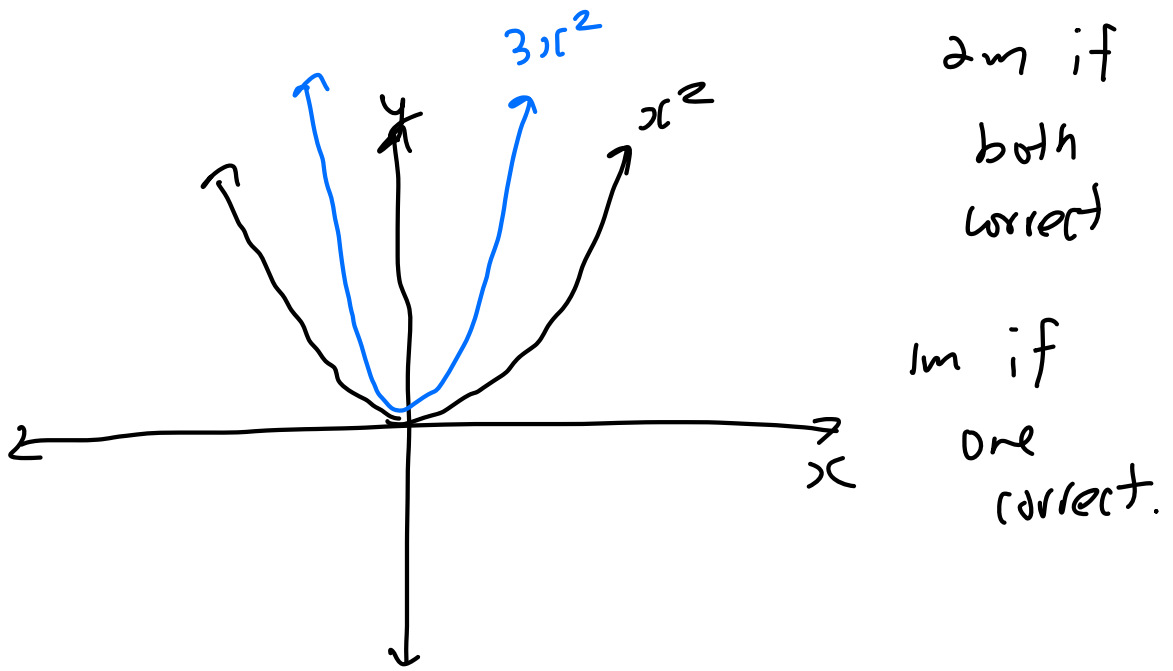
End of Question 14

Name _____

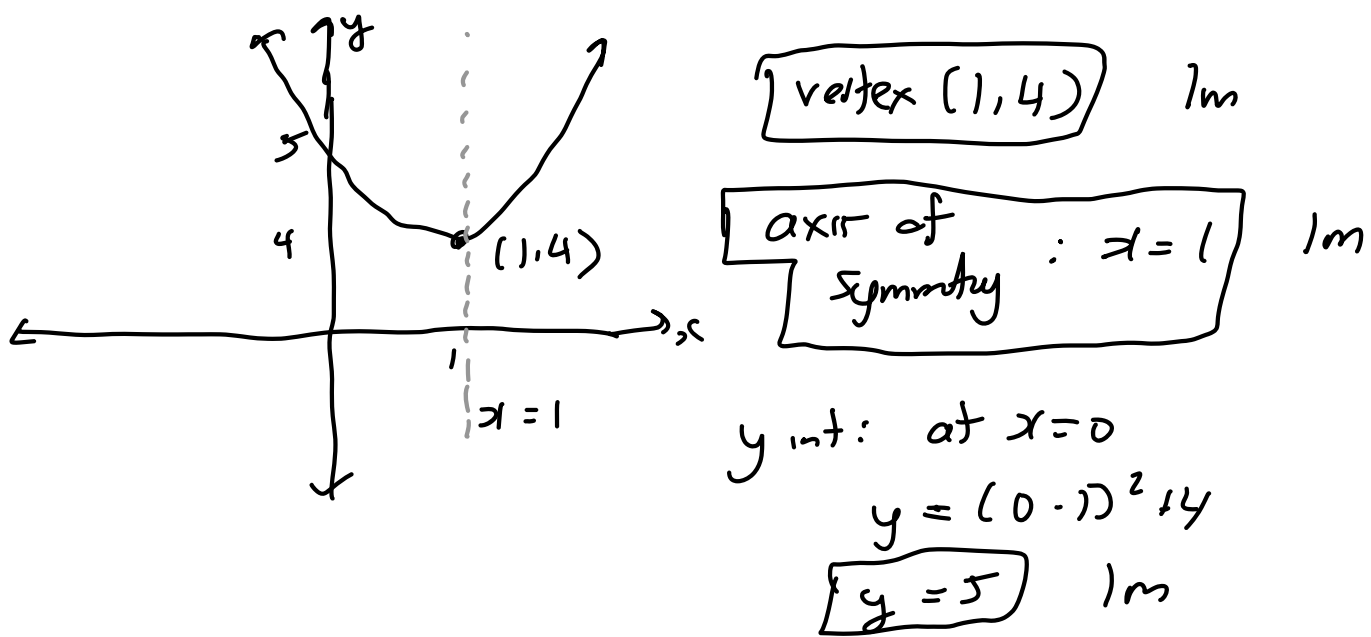
Question 15 (Algebra) (14 marks)

Marks

(a) Sketch $y = x^2$ and $y = 3x^2$ on the same set of axes. 2



(b) Sketch $y = (x - 1)^2 + 4$, showing the vertex, axis of symmetry and y-intercept. 4



1m for correct concavity

Question 15 continues on the next page

Name _____

Question 15 (continued)

Marks

(c) A parabola has the equation: $y = x^2 - 4x + 3$

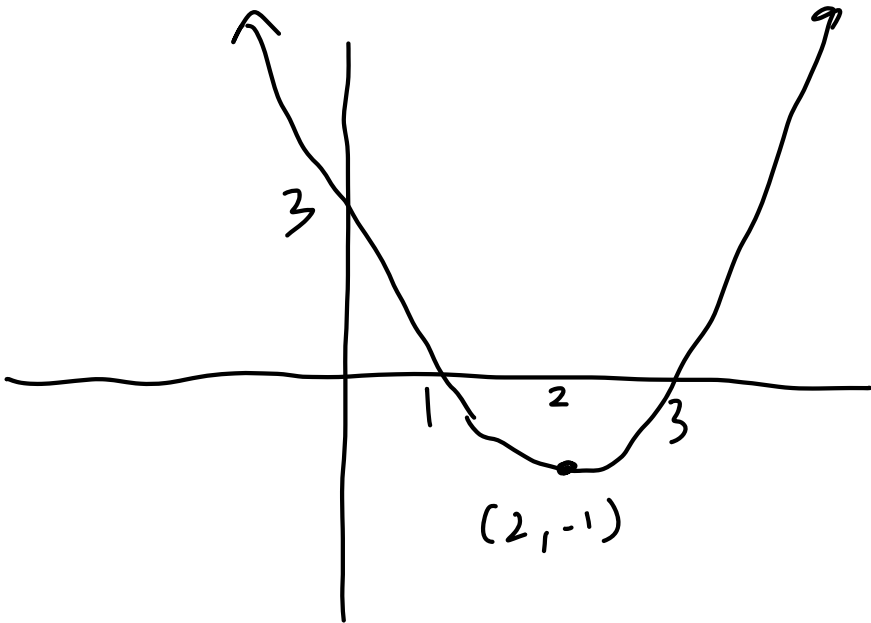
(i) Show that the x -intercepts for the parabola are at $x = 3$ and $x = 1$. 2

Im for finding x int at $y=0$
ie: $x^2 - 4x + 3 = 0$
Im for factoring $(x-3)(x-1) = 0$
& null factor: $\therefore x=3, x=1$

(ii) Find the y -intercept of the parabola. 1

at $x=0, y = 0 - 0 + 3 = \boxed{3}$

(ii) Sketch the parabola, showing the x -intercepts and y -intercept. 2



2m if
all features correct.

1m if only
some features correct.

Question 15 continues on the next page

Name _____

Question 15 (continued)

Marks

- (e) Hermione was on the phone trying to describe a parabola to Harry. 3
However, Hermione had bad reception and parts of the conversation had been cut out.
Hermione says that “the $y = x^2$ has been shifted _____ by 16 units”.
She also says, “the x-intercepts cuts at _____ and _____”.
Harry didn’t hear the exact x-intercepts but knew that **one** of the x-intercepts was between 2 and 4 (inclusive of 2 and 4).
The call then ended.

Given that the equation of a parabola is $y = a(x - h)^2 + c$, find the possible value(s) of a , h and c .

If you shift x^2 left or right, it only has ONE x intercept. But Φ says Two.
If you shift it up, there will be NO x int. $\therefore$ It's shifted down

$\therefore y = a(x - h)^2 - 16$ (Im for $c = -16$)
it isn't shifted left or right. $\therefore h = 0$
 $y = ax^2 - 16$ (Im for $h = 0$)

at $y = 0$,

$$ax^2 = 16$$

$$x^2 = \frac{16}{a} \rightarrow x = \pm \frac{4}{\sqrt{a}}$$

$$2 \leq \frac{4}{\sqrt{a}} \leq 4$$

$$\frac{1}{2} \geq \frac{\sqrt{a}}{4} \geq \frac{1}{4}$$

$$2 \geq \frac{1}{\sqrt{a}} \geq 1$$

$$4 \geq a \geq 1$$

$$\therefore 1 \leq a \leq 4$$

End of Question 15

End of Assessment Task



THE
KING'S
SCHOOL

2022

Year 9 Level 5.3
Assessment Task 4 (AT4)
November 2022

MULTIPLE CHOICE ANSWER SHEET

Select the alternative A, B, C or D that best answers the question and indicate your choice by filling in the appropriate response oval, completely.

					<i>Examiner's Use Only</i>	
1	☐ A	☐ B	☐ C	☐ D		
2	☐ A	☐ B	☐ C	☐ D	Number	/ 2
3	☐ A	☐ B	☐ C	☐ D		
4	☐ A	☐ B	☐ C	☐ D	Algebra	/ 2
5	☐ A	☐ B	☐ C	☐ D		
6	☐ A	☐ B	☐ C	☐ D		
7	☐ A	☐ B	☐ C	☐ D	Geometry	/ 3
8	☐ A	☐ B	☐ C	☐ D		
9	☐ A	☐ B	☐ C	☐ D		
10	☐ A	☐ B	☐ C	☐ D	Measurement	/ 3
					Total	/10

Name